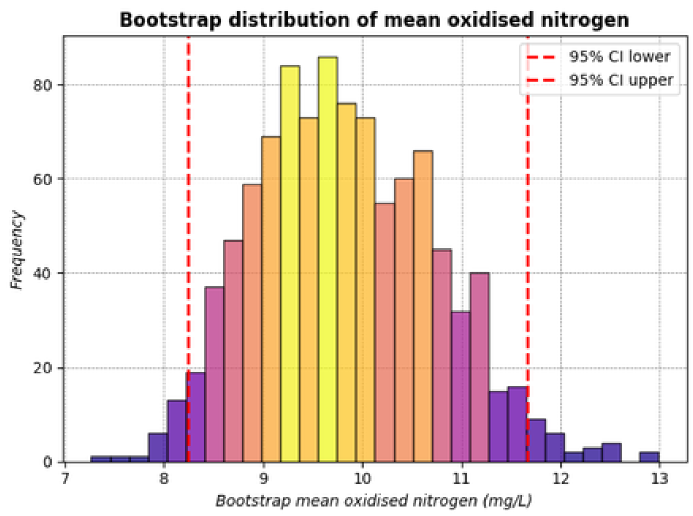
OXIDISED NITROGEN

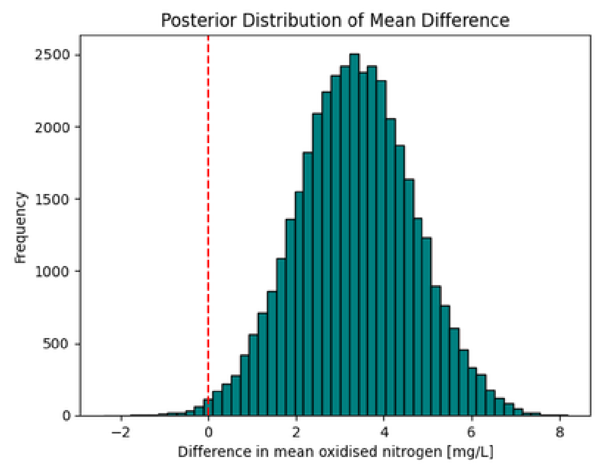
How have concentrations of oxidised nitrogen changed in Western Port Victoria both time and what are their spatial distributions?

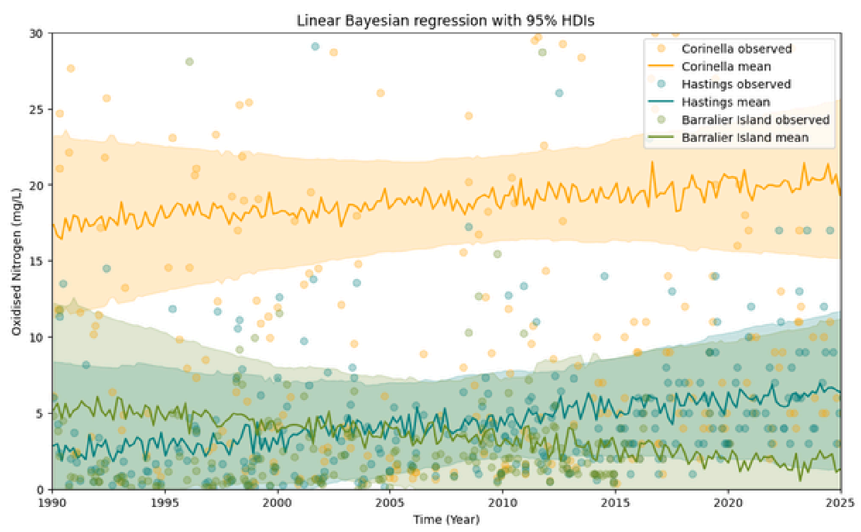
Introduction:
Western port is a large shallow embayment, which is It is surrounded by urban and agricultural lands, such that much of the natural landscape has been altered dramatically [3]. In my study of this location, I drew data forth from the 'EPA Western Port Water Quality Data 1990 - 2025' report to analyse how the local environmental had changed over time due to man-made factors.



Bootstrap:
Due to the relativly small amount of observations within the dataset, I decide to Bootstrap my data to create many more simulated datasets. Overall, when examining the individual bootstraps you can see a clear pattern where Corinella has a much higher mean than the others, illustrating a stronger concentration at the site

Hypothesis testing:
I want to examine if there has been a increase in mean NOx across the whole Western Port, from about the halfway point that the research was observed; 1990 - 2007, and 2007 - 2025. In calculating the probability, we found that the recent mean is > early mean, as we got 0.99 probability, an very strong likelihood.





Bayesian Regression:
The Corinella Site has a much higher % of NOx in comparison to the other sites. However what is fascinating is that Barralier Island actually decreases over time in comparison to Hastings which increases, crossing over each other around 2005. This disproves my hypothesis, indicating that there are others factors at play effecting the NOx content.

Conclusion:
Local infrastructure from nearby Corinella & Hastings has polluted the surrounding water quality. These factors actually explain why we see a decrease in NOx at Barrelier Island, as it off shore, next to the French Island National Park and thus less built up and can more easily recover.